

Analyzing Race Performance Dynamics in Formula 1 Using Race, Qualifying, and Weather Data

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1 Motivation

Formula 1 is a unique and highly complex sport where race outcomes are dictated by an intricate intersection of human skill, cutting-edge vehicle engineering, and unpredictable environmental factors. This project was motivated by a desire to mathematically untangle these elements and understand the true drivers of race-day performance by combining historical race, qualifying, and circuit data with race-day weather data.

Specifically, this analysis seeks to answer several core questions regarding race dynamics: How predictive is a driver's starting grid position of their final race result? Do specific weather conditions relate to changes between grid position and final finishing position? Furthermore, are certain circuits associated with greater race volatility than others, and do constructors differ in how often they outperform their starting position? Ultimately, the overarching question this project aims to answer through machine learning is: **which features are most useful for predicting whether a driver outperforms their starting position?**

Going into this analysis, the initial expectation was that while engineering (the constructor) and qualifying (the grid) are undeniably important, environmental factors, specifically rainfall and temperature shifts, would act as a massive equalizing force. The hypothesis was that wet weather would introduce high track volatility, neutralizing the advantages of top-tier machinery and allowing lower-tier teams to gain significant positions.

2 Data Source and Collection

To adequately capture the multifaceted nature of Formula 1, this project required synthesizing traditional race results with highly specific environmental data. The data collection phase was divided into extracting primary race data from a structured database and dynamically enriching it via an external weather API.

2.1 Primary F1 Data (The Ergast Database)

The foundation of the dataset was built using historical Formula 1 data from the Ergast database. To ensure the analysis remained relevant to modern F1 regulations and vehicle architectures, the scope was strictly filtered to cover the seven seasons from 2018 to 2024.

The raw data was initially distributed across six separate tabular CSV files: *races*, *circuits*, *drivers*, *constructors*, *qualifying*, and *results*. To construct a comprehensive master dataset suitable for machine learning, a robust relational merging process was executed. Using unique identifiers (such as `raceId`, `driverId`, and `constructorId`) as primary keys, the tables were iteratively joined. This transformation flattened the relational database into a single, comprehensive DataFrame where each row represented a specific driver's complete profile for a single race event.

2.2 External Weather Enrichment (Open-Meteo API)

Because environmental volatility was a core focus of the project proposal, historical weather conditions for each race had to be collected. This was achieved by integrat-

ing the Open-Meteo Historical Weather API.

To optimize the data pipeline and avoid redundant API calls for every driver, the master dataset was first aggregated to extract a list of unique race events. For each of the 149 unique races, the API was queried using the circuit’s exact geographical coordinates (latitude and longitude) and the specific date of the race. Four critical environmental variables were extracted for each event:

- Maximum Daily Temperature (°C)
- Mean Daily Temperature (°C)
- Total Precipitation (mm)
- Average Relative Humidity (%)

When querying the API, it was observed that for a small subset of races, the weather data was not being fetched from Open-Meteo. Given this instability when querying the external API iteratively, a robust fetching function was implemented featuring standard 10-second timeouts and exception handling to prevent pipeline failures.

2.3 Feature Engineering and Dataset Finalization

With the race and weather data successfully merged, custom target features were engineered to directly address the project’s research questions. The primary continuous feature, `positions_gained`, was calculated by subtracting a driver’s final official race position from their starting grid position. A negative value indicates positions lost, while a positive value indicates positions gained. For the machine learning phase, this was further abstracted into a binary target variable, `target_outperformed`, which classified whether a driver successfully advanced their position (1) or not (0).

Following the merging and feature engineering, a data cleaning pass was conducted. A small subset of observations (140 rows) was dropped due to missing API weather data for specific historical dates. The final dataset yielded 2,839 independent driver-race observations across 149 modern Formula 1 races, providing a rich foundation for exploratory data analysis, hypothesis testing and predictive modeling.

3 Data Analysis

The data analysis phase was structured sequentially to build a rigorous, mathematically sound foundation before applying complex predictive models. This progression moved from visual exploration to formal statistical hypothesis testing, and finally to machine learning classification.

3.1 Stage 1: Exploratory Data Analysis (EDA)

Before executing statistical tests, an extensive Exploratory Data Analysis (EDA) was conducted to understand the underlying distributions, identify baseline trends, and verify that the data met the assumptions required for parametric testing.

The first critical step was establishing the baseline predictability of a Formula 1 race. By plotting the average final finishing position against the starting grid position (Figure 1),

a clear linear trend emerged. Comparing the actual race outcomes against a theoretical "perfect predictability" baseline (where a driver finishes exactly where they start) visually confirmed the massive influence of qualifying performance.

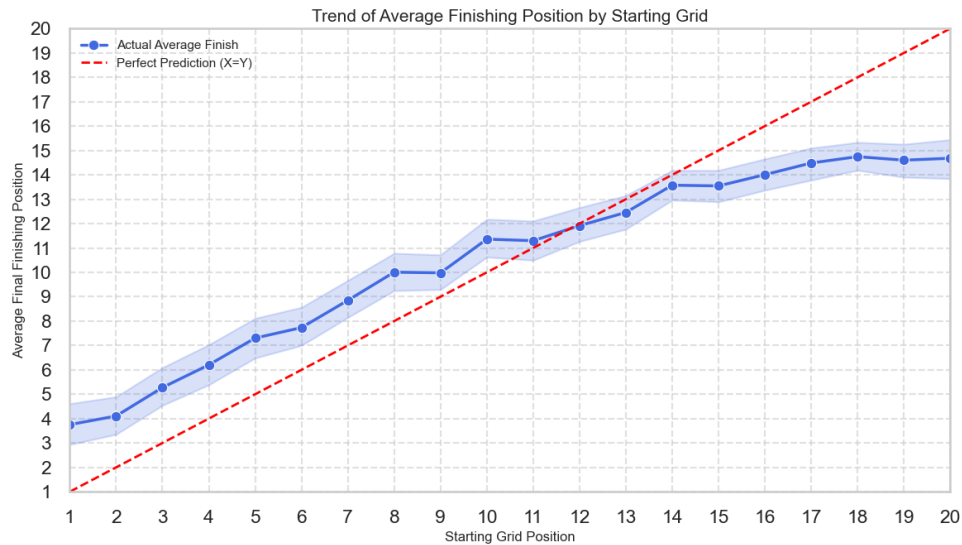


Figure 1: Trend of Average Finishing Position by Starting Grid. The dashed line represents perfect predictability ($X = Y$), while the solid line represents the actual historical average.

Next, the categorical impact of vehicle engineering was explored by mapping the average net positions gained/lost (our primary target variable) by each constructor (Figure 2). This horizontal bar chart immediately revealed a stark hierarchy within the sport. Teams such as Haas and Ferrari consistently averaged negative position changes (often qualifying high but falling back), while teams like Racing Point and Toro Rosso showed a historical tendency to slice through the field and gain positions.

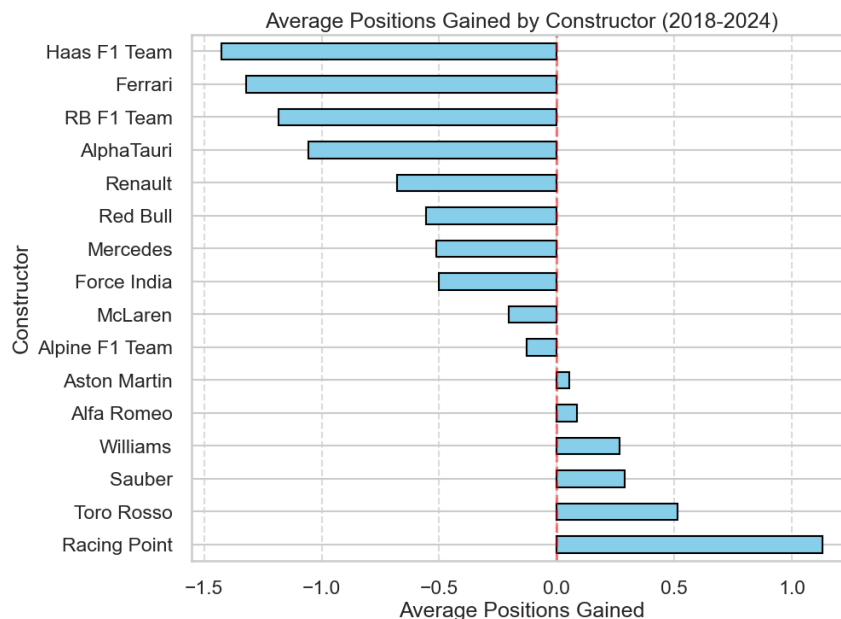


Figure 2: Average Positions Gained by Constructor (2018-2024). Teams on the right of the zero-line consistently move forward through the pack.

The distribution of the primary target variable (`positions_gained`) was also plotted across all races (Figure 3). The resulting histogram overlaid with a Kernel Density Estimate confirmed a roughly normal distribution centered near zero.

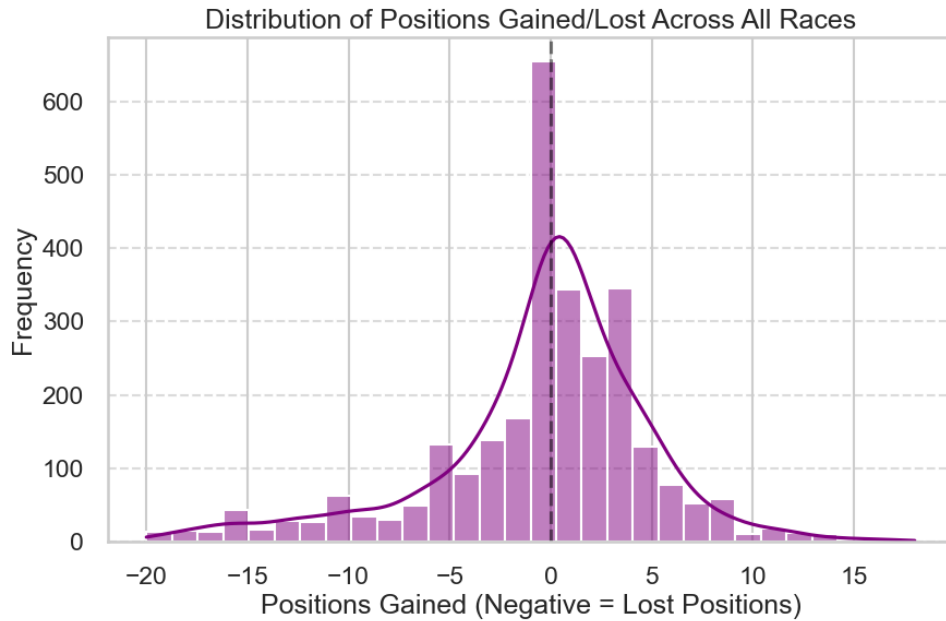


Figure 3: Average Position Gained/Lost in years 2018-2024

A subsequent correlation matrix was generated (Figure 4) to map the linear relationships between all numerical variables. Surprisingly, weak correlations were observed between weather metrics and position changes.

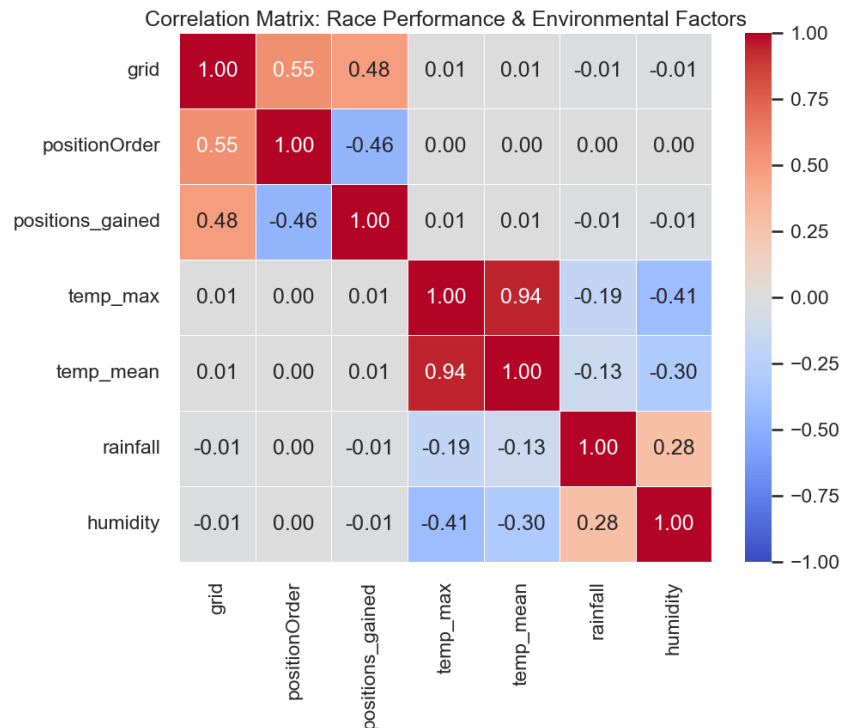


Figure 4: Correlation Matrix of Race Performance and Environmental Factors.

Additionally, to visualize the relationship between precipitation and volatility of race performance, two box plots were compared (Figure 5). The box plots appear to be similar visually. A slight difference in variance between dry and wet race conditions can be observed: a wider box and longer whiskers in the "Wet" category indicated slightly higher race volatility. This will be mathematically verified in the next subsection via Hypothesis tests.

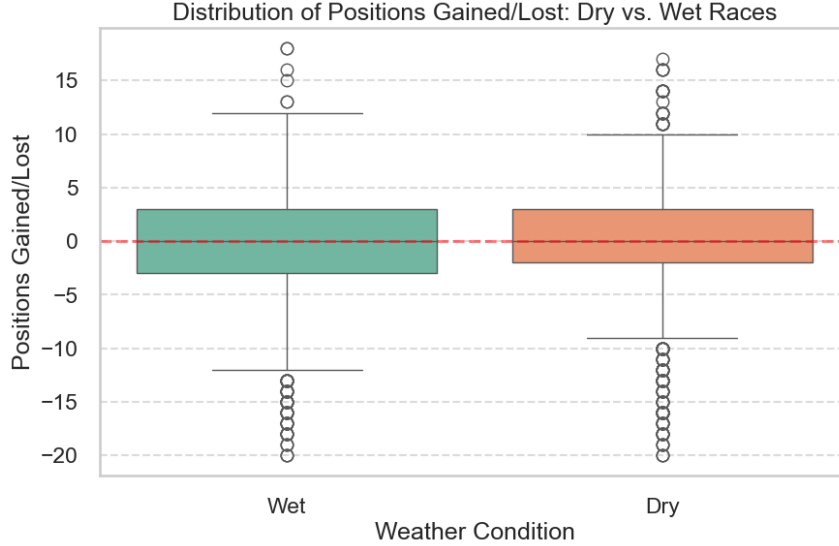


Figure 5: Box plots for positions gained in two distinct categories: Wet and Dry races.

3.2 Stage 2: Statistical Hypothesis Testing

While the Exploratory Data Analysis provided strong visual intuition, formal statistical testing was required to mathematically validate these relationships and eliminate subjective bias. To do this, four specific hypotheses were formulated to address the core questions of the project proposal. Each was tested using a standard significance level (α) of 0.05.

Test 1: Predictive Power of Qualifying (Two-Sample T-Test)

The first test sought to mathematically confirm the visual trend of grid dominance. A Two-Sample T-Test was utilized to determine if there was a significant difference in the average final race results between drivers who qualified in the top 10 versus those who qualified in the bottom 10. The test yielded a p-value approaching zero ($p = 5.48 \times 10^{-179}$). Consequently, the null hypothesis was rejected, confirming a massive, statistically significant difference in race outcomes based strictly on qualifying performance.

Test 2: Weather Volatility Impact (Two-Sample T-Test)

A core premise of this project was the assumption that adverse weather disrupts the normal race hierarchy. A second Two-Sample T-Test was deployed to test this by comparing the mean positions gained in strictly dry races versus wet races. Surprisingly, the test returned a p-value of 0.36. Because $p > 0.05$, the null hypothesis could not be rejected. Statistically, the presence of rainfall did not significantly alter the average number of positions a driver gained or lost, marking our first major divergence from the initial project expectations.

Test 3: Constructor Resiliency (Chi-Square Test of Independence)

To test the engineering factor, specifically whether a driver's ability to successfully outperform their starting grid position is dependent on the team they race for, a categorical Chi-Square Test of Independence was executed. A contingency table was created with the counts, mapping all constructors against a binary `outperformed_grid` variable yielded a p-value of 0.0001. The null hypothesis was rejected, proving a statistically significant dependency between a constructor and its historical resiliency in moving up the pack.

Test 4: Circuit Volatility (Chi-Square Test of Independence)

Finally, a second Chi-Square test was utilized to answer the proposal's question regarding track-specific chaos. The goal was to determine if certain circuits are inherently associated with higher race volatility (defined mathematically as instances where a driver gained or lost more than three positions). Returning a p-value of 0.0002, the test confirmed that extreme positional variance is highly dependent on the specific circuit layout, proving that not all tracks offer equal overtaking opportunities.

Key Takeaways

This statistical testing phase fundamentally reshaped the project's assumptions. While the tests confirmed the expected dominance of qualifying position, constructor engineering, and circuit layout, they debunked the hypothesis that weather acts as a primary driver of race volatility. The sport is less weather-dependent than hypothesized and this distinction shapes which features were prioritized in the subsequent modeling phase.

3.3 Stage 3: Data Preprocessing for Machine Learning

With the foundational statistical relationships established, the project transitioned into predictive modeling. The core objective of this phase was to build a binary classification model capable of predicting whether a driver would successfully outperform their starting grid position. Before any algorithm could be trained, the dataset required preprocessing to prepare the dataset for binary classification.

Target Variable Formulation

The continuous `positions_gained` feature was transformed into a strict binary target (y), labeled `target_outperformed`. Drivers who finished higher than they started (`positions_gained > 0`) were classified as 1 (True), while drivers who maintained their position or fell behind were classified as 0 (False).

Categorical Encoding and Multicollinearity

Machine learning algorithms inherently require numerical input. Consequently, nominal categorical variables, such as `constructor_name` and `circuit_name`, were transformed using dummy variable encoding. To maintain the statistical integrity of the models, particularly linear models like Logistic Regression, it was imperative to avoid the "dummy variable trap." By dropping the first encoded column for each category, the pipeline successfully mitigated perfect multicollinearity, ensuring that the model coefficients remained interpretable and independent.

Feature Standardization

The continuous numerical features within the dataset, such as starting `grid` and the weather metrics (`temp_max`, `rainfall`), were measured on vastly different scales. To prevent the models that rely on distance calculations from disproportionately weighting

variables, Z-score standardization (`StandardScaler`) was applied. This centered all numerical features around a mean of zero with a standard deviation of one, placing weather and grid variables on an equal mathematical playing field.

Feature Selection and Preventing Data Leakage

The most critical step in the pipeline preparation was the manual curation of the feature matrix (X) to prevent "data leakage." Data leakage occurs when a model is inadvertently trained on information from the future that it would not possess at the time of prediction.

To ensure the model was predicting reality rather than reverse-engineering the results, all hindsight variables were strictly dropped. This included final race time, points scored, laps completed, and fastest lap metrics. If left in the dataset, predicting grid advancement would be a trivial mathematical calculation, rendering the model useless for real-world pre-race prediction. Furthermore, identifiers (such as `raceId` or raw name strings) were dropped. The resulting feature matrix represented a pure, pre-race snapshot of conditions.

3.4 Stage 4: Model Validation and Selection

To ensure the machine learning results were robust and not the product of overfitting to a specific subset of the data, a 5-Fold Cross-Validation approach was utilized. The dataset was iteratively partitioned into five equal folds, training on four and testing on the unseen fifth.

A "model tournament" was conducted to evaluate four distinct classification algorithms across the cross-validation folds: Logistic Regression, k-Nearest Neighbors (kNN), Decision Trees, and Random Forests. Accuracy was selected as the primary baseline metric. Logistic Regression emerged as the champion model, achieving a cross-validated accuracy of 70.31%, outperforming the baseline Decision Tree (61.04%) and Random Forest (65.59%).

Due Diligence and Hyperparameter Tuning

Because Logistic Regression is a strictly linear model, due diligence required testing whether a complex, non-linear model could discover hidden patterns in the data if properly configured. To address the underperformance of the default Random Forest (likely caused by overfitting), hyperparameter tuning was performed using `RandomizedSearchCV`. By constraining the maximum tree depth and adjusting the minimum samples required to split a node, the optimized Random Forest was evaluated again. The tuned ensemble model achieved a maximum accuracy of 65.55%, still failing to surpass the Logistic Regression model. This suggests that F1 grid advancement is fundamentally a linear phenomenon, unbound by complex "if/else" environmental dependencies.

Final Evaluation and the Confusion Matrix

With Logistic Regression confirmed as the optimal algorithm, the model was trained on a standard 80/20 train-test split for a final evaluation. Beyond basic accuracy, the model achieved a robust ROC AUC score of 0.75, indicating a strong ability to distinguish between drivers who gain positions and those who do not.

The Confusion Matrix (Figure 6) revealed that the model is exceptionally strong at predicting True Negatives (identifying instances where a driver fails to improve their position). Examining the errors, the model produced more False Negatives (102) than False Positives (75). This indicates a slightly conservative bias in that the model inher-

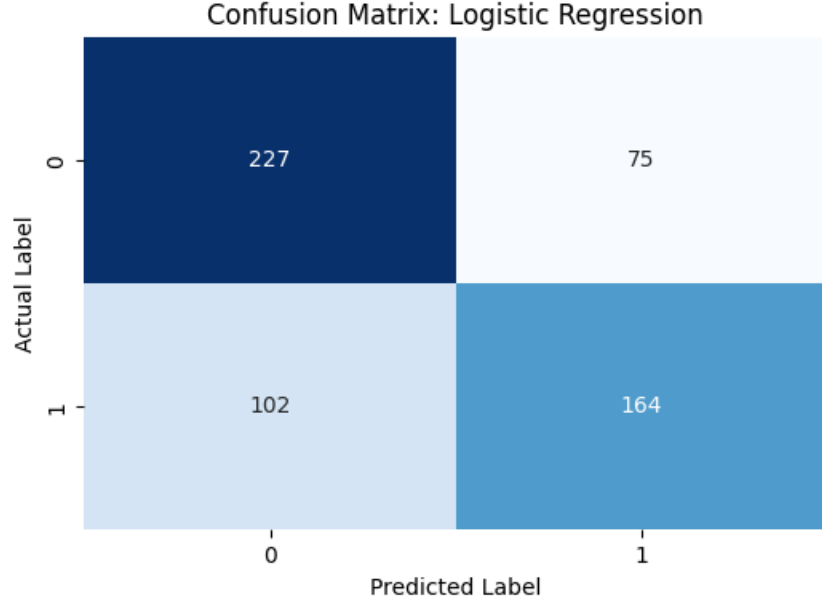


Figure 6: Confusion Matrix for the winning Logistic Regression model. The matrix highlights a high rate of True Negatives (227) and a slightly conservative bias regarding position gains.

ently assumes that overtaking is difficult and that maintaining position is the most likely outcome, which perfectly mirrors the physical reality of modern Formula 1 racing.

Feature Importance: Answering the Core Proposal Question

To definitively answer the project’s primary research question: which features are most predictive of race-day performance, the coefficients were extracted directly from the winning Logistic Regression model. Sorting these coefficients by their absolute values revealed the hierarchy of importance for gaining positions in Formula 1:

1. **Vehicle Dominance:** Constructor tier proved to be the ultimate predictor. Driving for top-tier teams like Mercedes (+1.76) or Red Bull (+1.54) yielded massive positive coefficients, while backmarker teams like Haas (-0.75) and Williams (-0.66) possessed strong negative coefficients.
2. **Grid Mathematics:** The starting grid position was highly predictive (+1.48). Drivers starting further back naturally possess higher statistical mobility to move forward, while those at the front have nowhere to go but backward.
3. **Circuit Bottlenecks:** Certain tracks were identified by the model as strong negative predictors. Meaning it is harder to advance positions there. Istanbul Park (-0.83) emerged as the strongest negative circuit predictor, closely followed by the Miami International Autodrome (-0.75) and the Hungaroring (-0.69). This mathematically confirms their real-world reputations as highly difficult circuits for overtaking.

4 Findings: Expectation vs. Reality

The ultimate value of this project lies in comparing the initial hypotheses outlined in the project proposal against the mathematical realities uncovered by the machine learning pipeline. The findings present a fascinating contrast between the perceived chaos of Formula 1 and its underlying, highly deterministic nature.

4.1 Expectations Matched: The Grid, The Car, and The Track

Going into the analysis, it was hypothesized that a driver's starting position, their constructor's engineering capabilities, and the specific circuit layout would heavily influence race outcomes. The data strongly confirmed these expectations:

- **The Absolute Dominance of the Grid:** The visual trend from the EDA was validated by the Two-Sample T-Test, proving that top 10 qualifiers finish significantly higher than the bottom 10. Furthermore, the machine learning pipeline identified the starting **grid** position as the second most powerful positive predictor (+1.48) of advancing positions. Statistically, the further back a driver starts, the higher their statistical mobility to move forward, while those at the front have nowhere to go but backward.
- **The Constructor Hierarchy:** The Chi-Square test proved a significant dependency between a team's engineering tier and their on-track performance. The Logistic Regression model quantified this: driving for a top-tier constructor like Mercedes (+1.76) or Red Bull (+1.54) yielded the most massive positive predictive coefficients in the entire dataset, whereas backmarker teams like Haas (-0.75) and Williams (-0.66) functioned as statistical anchors.
- **Circuit Bottlenecks:** The data confirmed that not all tracks are created equal. The Chi-Square test for track volatility proved that extreme positional variance is dependent on the circuit. The predictive model identified Istanbul Park (-0.83), the Miami International Autodrome (-0.75), and the Hungaroring (-0.69) as the strongest negative predictors, confirming their real-world reputations as notoriously difficult circuits for overtaking.

4.2 Expectations Subverted

- **The Weather Illusion:** The most significant finding of this project was the complete subversion of the primary working hypothesis. Initially, it was expected that environmental factors, specifically rainfall, temperature and humidity would act as an equalizing force, increasing track volatility in positions gained by the vehicles, weakening the advantages of top-tier machinery.

The reality proved to be different. The Two-Sample T-Test comparing dry races to wet races yielded a p-value of 0.36, indicating absolutely no statistically significant difference in the average number of positions gained or lost. The variance in wet races was identical to that of dry races.

Even more conclusively, when extracting the feature importance from the champion machine learning model, environmental factors failed entirely to breach the top 10 most predictive features. The data suggests a compelling pattern about

modern Formula 1: vehicle engineering and qualifying position vastly outweigh environmental volatility. Rain may cause individual driver errors, but it does not mathematically level the playing field or reliably help drivers outperform their grid positions.

- **The Fundamental Linearity of F1:** A secondary expectation was that predicting race outcomes would require a highly complex, non-linear machine learning algorithm (like a Random Forest) capable of identifying hidden "if/else" conditions (e.g. "if it is raining AND the driver is in a slow car, they will gain positions").

In reality, the optimized complex Random Forest model (65.55% accuracy) was outperformed by a simple, strictly linear Logistic Regression model (70.31% accuracy). This 5% performance gap serves as the final, overarching conclusion of the analysis: outperforming the grid in modern Formula 1 is fundamentally a linear phenomenon. A driver's mobility through the pack is bound almost entirely by a linear hierarchy dictated by the machinery they drive and where they qualify, rather than complex, hidden interactions with their environment.

5 Limitations and Future Work

While this project successfully identified the overarching linear dynamics of modern Formula 1 and quantified the predictive hierarchy of machinery and qualifying, the model is inherently bounded by the dataset. Acknowledging these limitations provides a clear roadmap for future iterations.

5.1 Limitations

- **Aggregated Weather Data (Low Temporal Resolution):** The integration of the Open-Meteo API provided daily weather aggregates, such as the maximum temperature and total daily rainfall. In reality, a track that begins wet but dries rapidly midway through the race creates a fundamentally different environment than a race with a constant light drizzle. In F1, massive position changes during wet races almost always occur during the chaotic "crossover" periods when teams gamble on pitting for different tire compounds. Because the dataset relies on a single daily aggregate, the model is completely blind to these mid-race environmental shifts. This lack of data resolution is likely the primary reason the statistical tests failed to find a correlation between weather and grid mobility.
- **The Driver vs. Constructor Confounding Variable:** A classic statistical challenge in motorsports analytics is separating the raw talent of the driver from the engineering dominance of the car. Since the algorithm assigns such massive mathematical weight to top-tier constructors, it effectively overshadows the human element. Consequently, the model struggles to predict outlier events where an exceptional driver outperforms the baseline capabilities of a bottom-tier car.
- **Mid-Race Chaos and Incidents:** The current predictive pipeline operates entirely on a pre-race snapshot (Grid, Constructor, Circuit, Weather). It is completely blind to in-race chaos, including Safety Cars, red flags, mechanical DNFs (Did Not Finish), first-lap collisions, and disastrous pit stops. Because these unpredictable

events heavily dictate final positions, a pre-race model will always possess an inherent accuracy ceiling.

5.2 Future Work

To address these limitations and improve the predictive power of the model, future iterations of this project should focus on enhancing feature engineering and data granularity.

- **Granular Weather Integration:** Instead of relying on daily weather aggregates, future work could improve the API extraction pipeline to pull hourly data. By matching the exact hour of the race start time to the hourly Open-Meteo data, the model could be fed a much more precise snapshot of the actual temperature and humidity at race start.
- **Incorporating Ergast Pit Stop Data:** While predicting mid-race chaos is difficult, the Ergast database does contain historical pit stop tables. A practical improvement would be to engineer new features from this data, such as the number of pit stops a driver took or their average pit lane time. This would allow the existing Logistic Regression and Random Forest models to account for more aspects of the sport.
- **Advanced Feature Engineering (Interaction Terms):** Since the simple linear model outperformed the complex ensemble model, future analysis could focus on creating manual interaction features for the Logistic Regression pipeline. For example, multiplying a weather variable by a constructor tier variable might mathematically expose how rain affects backmarker teams differently than front-runners, allowing a linear model to capture nuanced relationships it previously missed.
- **Multi-Class Classification:** Currently, the target variable is strictly binary (did the driver gain positions or not). A logical extension would be reframing this as a four-class problem: Lost Positions, Maintained, Gained 1–3, Gained 4+ and applying the same pipeline to predict not just whether a driver advances, but by how much.